# FORMALLY SMOOTH ON OBJECTS IS STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

*Editorial note.* Closing routine-check omission retained verbatim and flagged. Cross-chapter same-label note: stacks-more-morphisms L3111 / spaces-more-morphisms L4519 are morphism-level with Omitted proofs; hypotheses and conclusions differ, both count independently.

*Source reference.* Original equation 06D0: sources/stacks-criteria/pinned-original-source.tex, source lines 886–886. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $p : \mathcal{X} \to \mathcal{Y}$ and $q : \mathcal{Y} \to \mathcal{Z}$ be 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$. If $p$ and $q$ are formally smooth on objects, then so is the composition $q \circ p$.*

*Proof.* This is formal. Let $U \subset U'$ be a first order thickening of affine schemes over $S$, let $z'$ be an object of $\mathcal{Z}$ over $U'$, let $x$ be an object of $\mathcal{X}$ over $U$, and let $\gamma : q(p(x)) \to z'|_U$ be an isomorphism. As $q$ is formally smooth on objects there exist an object $y'$ of $\mathcal{Y}$ over $U'$, an isomorphism $\beta : y'|_U \to p(x)$, and an isomorphism $\gamma' : q(y') \to z'$ such that (06D0) is commutative. As $p$ is formally smooth on objects there exist an object $x'$ of $\mathcal{X}$ over $U'$, an isomorphism $\beta' : x'|_U \to x$, and an isomorphism $\gamma'' : p(x') \to y'$ such that (06D0) is commutative. The solution is to take $x'$ over $U'$ with isomorphism

$$q(p(x')) \xrightarrow{q(\gamma'')} q(y') \xrightarrow{\gamma'} z'$$

and isomorphism $\beta' : x'|_U \to x$. We omit the verification that (06D0) is commutative. □

*Date*: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.